

Investigating a Toroidal Foot Curvature Design in a 3D Passive Dynamic Walker

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Abstract—In recent years, there have been significant improvements in the naturalness of gait in humanoid and other legged robots, though many legged systems remain limited by their battery life. Research on passive dynamic walking aims to improve the energy efficiency of walking robots by drawing inspiration from the natural efficiency of human walking. Yet, there remains relatively little exploration of a crucial component of passive legged systems: the feet. In this study, we propose a toroidal foot curvature design for a passive walker and develop a modular robot with swappable feet to experimentally test variations in the foot curvature. We derive a prediction relating changes in the frontal radius of curvature to the rocking period, which our experimental results validate within 1% across all frontal radius pairs. We experimentally test variations of the frontal and the sagittal radii of curvature in our robot during its passive gait on a fixed slope and report distinct gait regimes characterized by increasing step length and walking speed for smaller radii or decreasing step length and walking speed for larger radii.

Keywords—Energy-efficient robotics, passive dynamic walking, bipedal locomotion, foot curvature, gait analysis, robot leg design.

I. INTRODUCTION

The utility of robots is evident in their ability to handle tasks in different areas with efficiency and precision, such as manufacturing, healthcare, and agriculture [1-5]. Within the robot family, legged robots offer superior mobility over complex and uneven terrain compared with wheeled robots, making them particularly well suited for tasks in challenging environments such as disaster zones, rocky terrain, and other unstructured areas [6-8]. However, legged robots' real-world implementations are limited by their complexity and energy inefficiency. For instance, there have been attempts to use bipedal legged robots to aid in planetary exploration, such as NASA's 2012 robot, Valkyrie, but a key limitation that prevented its implementation was its one-hour battery life [9]. A similar problem is present in legged robots that perform long-distance missions. A team of researchers designed a humanoid robot, the AgiBot 2, that walked across multiple terrains, setting the world record of 66 miles [10]. However, the AgiBot 2 required its battery to be replaced every 3 hours, making it an impractical long-term solution. These two robots, like many other legged systems in

use today, are highly actuated, meaning they rely on a large number of independently controlled motors to drive their motions [11]. A potential solution to the issue of energy inefficiency in legged robots is the utilization of passive dynamic walking. Passive dynamic walking is a mode of legged locomotion in which a robot walks down a slope using only its inherent mechanical design and gravity.

By investigating the key design features and mechanics that enable a robot to exhibit passive gait, researchers aim to make progress toward energy-efficient legged robots. The design of passive dynamic walkers has taken inspiration from natural human gait because, as observed through EMG studies by McMahon et al., a portion of human gait is unpowered by muscles, making it semi-passive [12]. Humans' energy-efficient walking is supported by the function of their feet. As investigated by Adamczyk et al., in human gait, the feet act analogously to arcs as the stance leg rolls over the foot [13].

While some passive dynamic walkers have been developed with point feet, they suffer from a small basin of attraction, or set of initial conditions from which passive walking could be achieved [14]. On the other hand, walkers developed with curved feet have exhibited elegant, anthropomorphic gaits [11, 14, 15]. Ruian et al. utilized curved feet and even had knees and counter-swinging arms [15]. During the development of that relatively complex model, simulations were unable to identify a design that produced a stable gait. Instead, the successful design was achieved through iterative experimentation.

Addressing the challenge of simulating passive walkers with curved feet, Tehrani recently made a breakthrough by solving the coupling between frontal and sagittal planes [16]. However, the simulation of his model with spherical feet was unable to find a set of design parameters that yield periodic gait, and did not account for some of the additional dynamics present in a physical passive walker, such as the yaw motion. Overall, the mechanics and potential of passive dynamic walking are not yet fully understood, and the accurate simulation of fabricable models remains a notable technical challenge. Given these limitations of simulation-based approaches, physical experimentation offers a valuable complementary path for exploring curved foot geometries.

In this work, we experimentally investigate a toroidal foot curvature design and its capability to modulate the behavior of a 3D Passive Walker. In Section II, we derive a relationship between frontal plane curvature and rocking period. In Section III, we develop and test a modular robot with swappable feet that

can walk passively down an 8-foot shallow slope. In Section IV, we present the results of our experimental variations of foot curvature in frontal and sagittal planes, and in Section V we summarize this study and discuss future work.

II. METHODOLOGY

A. Model of Passive Walking

Our passive walker design, inspired by the model introduced by Tedrake et al. [17], consists of two legs connected by a pin joint at the hip, and toroidal curved feet that provide foot clearance by lateral rocking, as shown in Fig. 1. The foot surface profile is defined by Eq. (1), where x and y denote local coordinates in the foot's local frame along the frontal and sagittal directions, respectively, R_f is the radius of curvature in the frontal plane, and R_s is the radius of curvature in the sagittal plane.

$$z(x, y) = \sqrt{\left((R_f - R_s) + \sqrt{R_s^2 - y^2}\right)^2 - x^2} - R_f \quad (1)$$

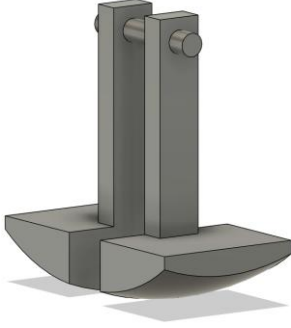


Fig. 1. Visual Render of Model Description.

The model walks down a shallow slope by rocking laterally onto one stance leg, lifting the opposite leg off the ground and allowing it to swing forward. As the walker returns to an upright position, the swing leg contacts the ground and becomes the new stance leg, initiating the next step with the opposite leg. This alternating sequence of stance and swing phases enables the robot to descend the slope, during which gravitational potential energy is converted into kinetic energy, while energy is dissipated through foot-ground collisions and friction.

B. Theoretical Framework

The toroidal foot curvature design allows us to modify the radius of curvature in the frontal plane independently from the sagittal plane. As shown in Fig. 2, the walker rocks laterally on its frontal-plane curvature and rolls forward during the stepping motion over its sagittal-plane curvature. Since solving the coupled 3D dynamics between the frontal and sagittal planes in passive gait with non-spherical feet remains an open technical challenge, this section focuses on the *timing* of walking generated by the oscillating rocking motion, which is characterized by the rocking period. To derive such a relationship, we model rocking on a level surface and investigate how R_f affects the rocking period using energy conservation between foot collisions. In Section V, our approximation is

tested against the rocking period during the gait cycle for varied R_f , where energy is provided by the slope.

In this derivation, we assume that the feet roll without slipping on the ground, that the walker is a single rigid body, that energy is conserved between collisions, and that the center of mass (COM) is below the center of the circle of curvature in the frontal plane for a statically stable configuration.

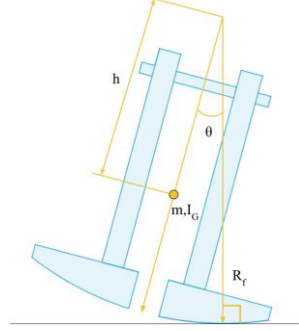


Fig. 2. Frontal Plane Model of Passive Walker.

As the walker rocks by rolling over the foot, its COM moves because of both the walker's rotation and its translation. The location of the center of mass can be modeled as follows:

$$x_G = R_f \theta - h \sin \theta, \quad y_G = R_f - h \cos \theta \quad (2)$$

where x_G and y_G are the horizontal and vertical positions of the center of mass, h is the distance from the foot's center of curvature to the center of mass, and θ is the rocking angle measured from vertical.

To determine the walker's total mechanical energy, we first compute the velocity of its COM by differentiating its position with respect to time. We then calculate the corresponding kinetic and potential energies:

$$\dot{x}_G = (R_f - h \cos \theta) \dot{\theta} \quad (3)$$

$$\dot{y}_G = (h \sin \theta) \dot{\theta} \quad (4)$$

$$V = mg(R_f - h \cos \theta) \quad (5)$$

$$K = \frac{1}{2} [I_G + m(R_f^2 + h^2 - 2R_f h \cos \theta)] \dot{\theta}^2 \quad (6)$$

where I_G is the walker's resistance to spinning about its center of mass. Since the normal force and friction act at the contact point, which has zero velocity relative to the ground during rolling without slipping, both forces do no work. Gravity is therefore the only force doing work on the system, and the total energy $E = K + V$ remains constant throughout the smooth rocking motion.

To yield linear equations of motion, small-angle approximations ($\cos \theta \approx 1 - \frac{\theta^2}{2}$) for potential energy and ($\cos \theta \approx 1$) for rotational inertia are applied near $\theta = 0$. The lateral tilt angle is physically bounded by the foot's physical arc length, yielding a maximum tilt angle of $\theta_{max} = 0.452$ radians, which is calculated using the smallest R_f experimentally tested. At this maximum tilt, the second-order cosine approximation applied to Eq. (5) yields a relative truncation error of 1.72% in

the change in gravitational potential energy, supporting its use in the linearization of the potential-energy term.

By retaining terms to the order required for a linear equation of motion, this reduces the potential energy to a parabola in θ and reduces the inertia term to a constant:

$$V \approx mg \left[R_f - h \left(1 - \frac{\theta^2}{2} \right) \right] = mg(R_f - h) + \frac{1}{2}mgh\theta^2 \quad (7)$$

Let ($h_0 = R_f - h$) denote the walker's standing height above the ground:

$$V \approx mgh_0 + \frac{1}{2}mg(R_f - h_0)\theta^2 \quad (8)$$

Since $\cos\theta$ appears only in the coefficient of $\dot{\theta}^2$ in Eq. (6), $\cos\theta \approx 1$ suffices here, giving:

$$K \approx \frac{1}{2} \left[I_G + m(R_f - h)^2 \right] \dot{\theta}^2 \quad (9)$$

The moment of inertia in Eq. (9) can be expressed using the parallel-axis theorem as

$$I_C = I_G + mh_0^2 \quad (10)$$

where I_C is the moment of inertia about the foot-ground contact point.

$$E = \frac{1}{2}I_C\dot{\theta}^2 + \frac{1}{2}mg(R_f - h_0)\theta^2 + mgh_0 \quad (11)$$

Since E is conserved, $\frac{dE}{dt} = 0$:

$$I_C\dot{\theta}\ddot{\theta} + mg(R_f - h_0)\theta\dot{\theta} = 0 \quad (12)$$

Canceling the common factor $\dot{\theta}$ gives:

$$\ddot{\theta} + \frac{mg(R_f - h_0)}{I_C}\theta = 0 \quad (13)$$

This yields a period of

$$T = 2\pi \sqrt{\frac{I_C}{mg(R_f - h_0)}} \quad (14)$$

By comparing two frontal radii, R_1 and R_2 , on otherwise identical robots where m , I_C , and h_0 are unchanged, we get:

$$\frac{T_2}{T_1} = \sqrt{\frac{R_1 - h_0}{R_2 - h_0}} \quad (15)$$

As shown in Eq. (15), $R_2 > R_1$ predicts $T_2 < T_1$, meaning that a larger radius of curvature in the frontal plane produces a shorter rocking period.

This derivation approximately captures the relationship between R_f and the rocking period. In the following sections, we develop an experimental passive dynamic walker to analyze its behavior across gait cycles. In testing, we evaluate our approximation against the rocking period data we collect. We also report empirical data on the effects of changing frontal and sagittal curvatures on gait behavior more broadly, which is not captured by existing theoretical models, including Tehrani's, which is limited to spherical feet with a single radius of curvature [17].

III. EXPERIMENTAL DESIGN

As shown in Fig. 3, our walker consists of 3D-printed legs and feet that were designed in the Fusion CAD software. Design considerations for the walker body were guided by physical experimentation and informed by insights from previously developed passive dynamic walkers.

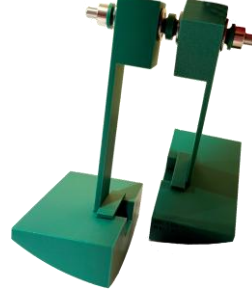


Fig. 3. The designed walker in this study.

According to McGeer, friction at the hip joint should be minimal and consistent, as damping causes the walk to decay if energy losses are not balanced out by the slope's gravitational input [14]. To achieve this, ball bearings are used along with 3D-printed inner bearing spacers and shaft collars.

Physical parameters, specifically proportions in addition to environmental influences such as the slope and surface, were calibrated experimentally until the walker displayed a reasonably stable gait characterized by consistent step length, period, and a straight trajectory. Beyond the walker's total energy dynamics, a major source of instability and gait irregularity was yaw motion, or twisting about the vertical axis. To experimentally resolve this issue, Collins et al. used counter-swinging arms for their relatively large robot. However, when scaling down our model, we found that yaw was reduced enough for us to use a scrubbing torque as a solution, which we achieved by covering the wooden slope in a textured surface. The textured material reduced yaw enough to avoid increasing the mechanical design's complexity with the counter-swinging arms.

For the 3D-printed feet, we designed a swappable foot mechanism to test various foot types on the same body (Fig. 4). To isolate the toroidal curvature, the feet's geometric variables, such as length, width, and walker height contribution, were kept constant. It is important that the foot be long enough to avoid rolling onto the foot's edge during gait. In addition, the COM was not altered by ensuring that each set of feet had a consistent mass of approximately 27.45 ± 0.26 g, achieved by adjusting the infill density.

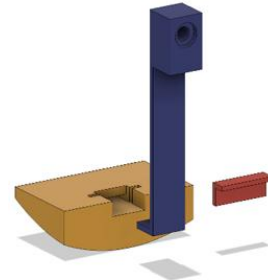


Fig. 4. Modular robot model with swappable feet.

Incorporating these design considerations, a baseline walker model was developed in which passive walking could be achieved by tuning the foot curvature. Started by hand, as done in Fig. 5, it could walk relatively consistently for approximately 40 steps down our 8-foot ramp at 0.0663 rad, or 3.80 degrees (8-foot walk.mov). The walker weighs 450.9 g and has a leg length of 12.87 cm. The center of mass height, h_0 , is approximately 61.5 mm. The feet in the 8-foot walk have an R_f of 120 mm and an R_s of 100 mm.



Fig. 5. Physical passive walker on inclined ramp.

From this design, only the curvatures of the feet are modified. Each configuration is tested for 3 trials, wherein rocking period, step length, step count, walking speed, and standard deviations are measured to characterize and evaluate the walking cycle. Measurements are conducted via video analysis at 4K 60 fps with two camera angles located above and on the side of the walker.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Rocking Period Results and Analysis

R_f was tested at 110 mm, 120 mm, 130 mm, and 140 mm, while R_s was held constant at 100 mm. As R_f increased from 110 mm to 140 mm, the rocking period decreased as shown in Fig. 6.

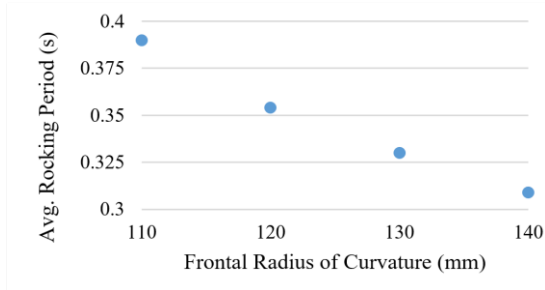


Fig. 6. Increasing R_f decreases rocking period.

As predicted in Eq. (15), an increase in R_f should result in a decrease in rocking period, which is supported by the result in Figure 5. The low standard deviation of the rocking period, as shown in Table I, reveals that the rocking motion acts as a simple harmonic oscillation with a relatively constant period, despite the possibility of varying amplitudes.

Table I. Standard deviation of rocking periods across R_f per trial

R_f (mm)	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)
110	0.02	0.02	0.01

120	0.01	0.01	0.02
130	0.02	0.02	0.02
140	0.03	0.02	0.02

To further compare our prediction with the experiment, the predicted ratio, as given in Eq. (15), is calculated for all possible pairings among the four frontal radii tested and is compared with the measured periods (Table I). Across the six pairings, predicted and measured ratios differ by at most 1.0%.

Table II. Predicted vs. measured ratio of rocking periods across frontal radii pairs

Radii (mm)	Predicted Ratio	Measured Ratio	Difference
110 vs 120	0.910	0.908	0.2%
110 vs 130	0.841	0.846	0.5%
110 vs 140	0.786	0.792	0.6%
120 vs 130	0.924	0.932	0.9%
120 vs 140	0.863	0.873	1.0%
130 vs 140	0.934	0.936	0.2%

This result is noteworthy because the theoretical framework assumes smooth rocking on level ground, while gait introduces foot-ground collisions, yaw motion, and a time-varying inertia contribution from the swing leg. Despite these additional dynamics, the measured period ratios differ from the theoretical predictions by at most 1.0%, suggesting that the unmodeled dynamics do not substantially alter the relationship between R_f and the rocking period over the tested conditions.

Furthermore, while keeping R_f fixed at 120 mm, R_s was tested at 80 mm, 100 mm, and 120 mm. The rocking period was measured across three trials for each configuration, and it was observed that R_s has a minimal effect on the rocking period during gait, as shown in Table III. However, due to the lack of currently existing mathematical explanations of coupling effects, no conclusion can be decisively drawn as to whether R_s does in fact affect the rocking period.

Table III. Intra-Trial and Inter-Trial rocking period standard deviations across R_s values

R_s (mm)	Mean Rocking Period (s)	Intra-Trial SD (s)	Inter-Trial SD (s)
80	0.36	0.01	0.01
100	0.35	0.01	0.02
120	0.34	0.01	0.01

B. Passive Gait Behaviors

The walker's toroidal foot curvature was modified in frontal and sagittal planes independently, and was tested on a fixed slope. The gait behavior was characterized with step length and speed at small and large R_f values while R_s was kept constant at 100 mm. Fig. 7 reveals different speed progressions for the R_f tested, indicated by the positive and negative speed gradients for the 110 mm and 140 mm frontal radii, respectively. In Fig. 8, an increasing step length gradient is present for the 110 mm R_f as evidenced by the positive slope of the trend line, and a decreasing step length gradient is present

in the gait of the walker with the 140 mm R_f . These distinct behavioral regimes can be explained by the balance between energy gained from the slope and energy dissipated during walking. The accelerating gait pattern at 110 mm R_f has an energy gain from the slope greater than the energy it loses from damping, characterized by steps that get longer and faster until the walker topples forward. The decaying gait pattern at 140 mm R_f has an energy deficit as it loses more energy due to damping than it gains from the slope, characterized by steps that get shorter and slower until the walker slows down to a stop. We propose this energy balance as a logical interpretation of the observed data. Future advancements in theoretical models may capture these results.

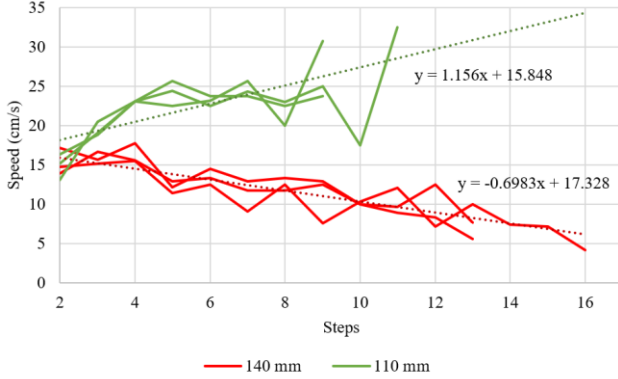


Fig. 7. Speed decreases during gait across trials for $R_f = 140$ mm; speed increases during gait across trials for $R_f = 110$ mm.

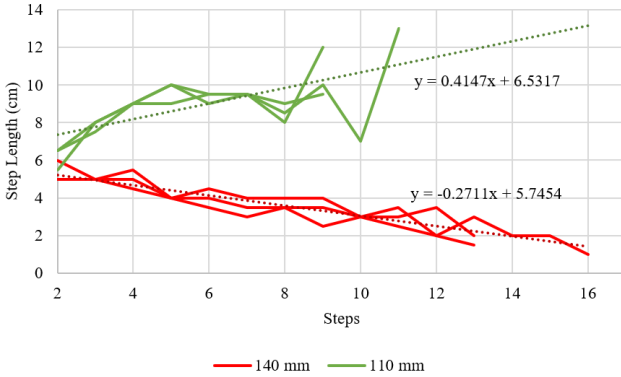


Fig. 8. Step length decreases during gait across trials for $R_f = 140$ mm; step length increases during gait across trials for $R_f = 110$ mm.

The walker takes relatively fewer steps with the 110 mm R_f because as it walks it speeds up and takes progressively longer steps until it falls over, while in comparison, the walker with the 140 mm R_f is able to slow down to a shuffle until finally coming to a stop on the slope due to its statically stable configuration.

A similar pattern is observed when testing R_s at 80 mm and 120 mm while keeping R_f fixed at 120 mm, as shown in Fig. 9 and Fig. 10. Overall, these results show that the gait behavior of a passive dynamic walker can be modulated with a toroidal foot curvature design.

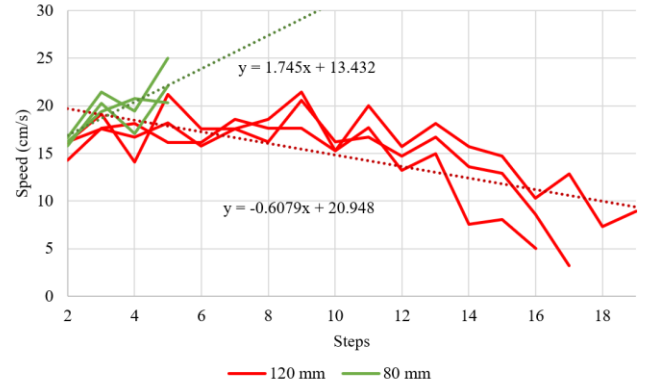


Fig. 9. Speed decreases during gait across trials for $R_s = 120$ mm; speed increases during gait across trials for $R_s = 80$ mm.

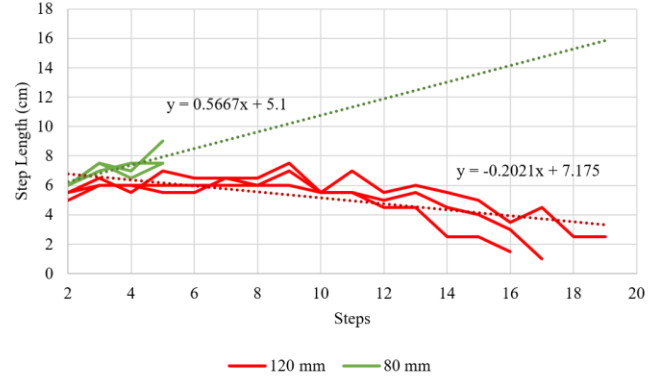


Fig. 10. Step length decreases during gait across trials for $R_s = 120$ mm; step length increases during gait across trials for $R_s = 80$ mm.

Downstream of this research, toroidal foot curvature in passive walking may provide applications for humans. Adamczyk et al. have shown that attaching rigid arcs to the bottom of a person's shoes can make their walking more energy-efficient by taking advantage of humans' natural passive mechanics [13]. Studying how foot curvature in the frontal and sagittal planes affects gait behavior may lead to foot curvature designs that aid in human gait by helping guide natural walking behaviors for those with injuries or disabilities that cause unnatural gait patterns.

V. CONCLUSIONS AND FUTURE WORK

To explore a toroidal foot curvature design in a passive walker, we presented a closed-form small-angle approximation relating the frontal radius of curvature to the rocking period, and validated it experimentally on a physical passive walker with swappable feet. The predicted and measured rocking periods were within 1.0% across all combinations of tested frontal radii despite the mathematical prediction neglecting swing-leg inertia coupling, energy losses from foot collisions, and yaw motion. Changes in the sagittal radius of curvature produced a minor effect on the rocking period, but it could not be decisively concluded that the sagittal radius of curvature does not affect the rocking period. Experimentally, we found that varying either frontal or sagittal radii affected the walker's gait behavior, where small and large radii produced positive and negative step length and walking speed gradients, respectively.

This work was limited by the variety of information that could be captured to characterize the gait. The video-based data collection method, using two camera angles, could discern foot collisions to measure rocking period and step length, but was not able to precisely measure angles. Future work includes developing a sensor-based data acquisition system to provide more comprehensive measurements, including the inter-leg angle at heel strike and the walker's yaw motion during the gait cycle. Additionally, this work could be furthered by developing a theoretical model to capture the coupling between frontal and sagittal planes in a passive walker with a toroidal foot geometry, as well as incorporating knees into the walker to explore a more anthropomorphic design.

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